

VELLORE INSTITUTE OF TECHNOLOGY

VIT VELLORE

School of Computer Science and Engineering (SCOPE)

Database Systems – BACSE202

PRODUCT ORDER MANAGEMENT SYSTEM

A Data Model and Entity-Relationship Design Assessment

Submitted by

ABHAY KUMAR

Registration Number: 25BCE2537

Vellore Institute of Technology, Vellore Campus, Tamil Nadu, India

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Abstract

This assessment presents the complete data model and entity-relationship design of a Product Order Management System, developed as part of the Database Management Systems coursework at VIT Vellore. The system is designed to help a retail/e-commerce business track customers, suppliers, and the products they list, capture every order a customer places along with the individual line items within it, and record the payments and shipments that settle and fulfil each order, while also allowing customers to raise post-sale returns and reviews against what they purchased.

The design is documented here through eight interlinked tables, an entity-relationship diagram showing every primary and foreign key, and the underlying spreadsheet data model, which is made directly accessible alongside the explanatory text through the links provided in this report.

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1. Introduction

Online and in-store retail has grown increasingly dependent on digital order management, and modern retailers process orders every day across websites, mobile apps, and marketplaces. A Product Order Management System is the layer of software responsible for recording what a customer bought, from whom the product was sourced, how it was paid for, how it was shipped, and what happened afterwards if the customer returned it or left a review. This report documents the design of such a system at the data level: what information it stores, how that information is organised into tables, and how those tables relate to one another.

The system modelled here follows one guiding principle — every item that leaves the warehouse should be traceable back to a customer, an order, a supplier, and a payment. Meeting that principle requires more than a single orders table; it requires a connected model of customers, suppliers, products, orders, order items, payments, shipments, and returns/reviews, each of which is described in the sections that follow.

2. Objectives

- To design a normalised, multi-table data model capable of representing customers, suppliers, products, orders, order items, payments, shipments, and returns/reviews.
- To identify and clearly mark the primary key and foreign key of every table, so that referential relationships between tables are unambiguous.
- To represent the complete schema visually through an entity-relationship diagram in Chen notation.
- To populate the model with representative sample data and make that data directly accessible alongside the written report.
- To discuss the practical applications, advantages, and possible future enhancements of the system.

3. System Overview

At the centre of the model sits the Orders table, since every other transactional table exists either to describe an order's contents or to react to it. An order is always placed by a customer and, through the associative Order Items table, is made up of one or more products, each carried at its own quantity and unit price. Products are sourced from Suppliers independently of any particular order, so that stock levels and supplier relationships can be managed without touching historical order records. Once an order is confirmed, it is settled by a row in Payments and fulfilled by a row in Shipments, both of which reference the order through a foreign key and can progress through their own status values independently of the order's own status.

Customers sit at the top of the hierarchy: every order and every return or review is registered to exactly one customer. Returns & Reviews closes the loop after fulfilment — a customer can raise a cancellation, a damage report, or a review against a specific order and the specific product within it, which keeps post-sale feedback traceable back to both the transaction and the product concerned.

4. Data Model — Table-by-Table Description

The schema is composed of eight tables. Each table's primary key (PK) uniquely identifies its own rows, and its foreign keys (FK) point to the primary key of a related table, which is how the tables are linked together into a single connected model.

4.1 Customers

Customers is the root of the system — every order and every return/review reference eventually traces back to a row here.

Attribute	Key	Usage / Purpose
Customer_Number [PK]	PK	Uniquely identifies each customer; every other table that refers to a person points back to this value.
Full_Name	-	The customer's name, used for identification throughout the system.
Email_Address	-	Primary contact channel, used for order and shipment notifications.
Phone_Number	-	Secondary contact channel used for delivery coordination.
City	-	The customer's home city, used for regional reporting and default delivery context.
Joined_On	-	The date the customer's relationship with the store began.
Standing	-	Current profile state (Active or Inactive).

4.2 Suppliers

Suppliers records every vendor that stocks the store with products. Supplier standing feeds directly into purchasing decisions.

Attribute	Key	Usage / Purpose
Supplier_Number [PK]	PK	Uniquely identifies each supplier.
Supplier_Name	-	The vendor's business/display name.
Contact_Person	-	The named point of contact at the supplier.
Contact_Email	-	Channel used for purchase-order and stock communication.
City	-	Where the supplier operates from.
Working_With_Us_Since	-	The date the supplier relationship began.
Standing	-	Current relationship state (Active or On Hold).

4.3 Products

Products records the store's catalogue. Every product belongs to exactly one supplier and carries the stock currently on hand.

Attribute	Key	Usage / Purpose
Product_Number [PK]	PK	Uniquely identifies each product.
Product_Name	-	The display name of the product shown to customers.
Supplied_By [FK]	FK	Points to the owning row in Suppliers.
Category	-	The product's classification, e.g. Home & Living, Kitchen, Electronics.
Price	-	The current list price of the product.
Currency	-	The currency the price is denominated in.
Stock_On_Hand	-	The quantity currently available in inventory.
Listed_On	-	The date the product was first listed for sale.

4.4 Orders

Orders is the centre of the model — every other transactional table exists to describe, settle, fulfil, or react to a row here.

Attribute	Key	Usage / Purpose
Order_Number [PK]	PK	Uniquely identifies each order.
Placed_By [FK]	FK	Points to the Customers row that placed the order.
Order_Date	-	The date the order was placed.
Delivery_Address	-	The address the order is to be delivered to.
Order_Total (derived)	-	The total value of the order, derived by summing the Line_Total of every related Order Items row.
Currency	-	The currency the order total is denominated in.
Order_Status	-	Processing, Shipped, Delivered, or Cancelled.

4.5 Order Items

Order Items is the associative table that resolves the many-to-many relationship between Orders and Products, recording exactly which products, at what quantity and price, made up a given order.

Attribute	Key	Usage / Purpose
Item_Reference [PK]	PK	Uniquely identifies each order line item.
Belongs_To_Order [FK]	FK	Points to the Orders row this line item is part of.
Product_Purchased [FK]	FK	Points to the Products row that was purchased.
Quantity	-	The number of units of the product purchased in this line item.
Price_Each	-	The unit price at the time of purchase, captured independently of the current product price.
Line_Total (derived)	-	Quantity multiplied by Price_Each; rolls up into the parent order's Order_Total.

4.6 Payments

Payments records how and when each order was settled, independently of the order's own status.

Attribute	Key	Usage / Purpose
Payment_Reference [PK]	PK	Uniquely identifies each payment.
For_Order [FK]	FK	Points to the Orders row this payment settles.
Amount_Paid	-	The amount received from the customer.
Currency	-	The currency the payment was made in.
Payment_Method	-	UPI, Credit Card, Net Banking, and similar.
Paid_On	-	The date the payment was captured.
Payment_Status	-	Successful, Refunded, or similar states.

4.7 Shipments

Shipments records how each order physically reaches the customer, independently of the order's own status.

Attribute	Key	Usage / Purpose
Shipment_Reference [PK]	PK	Uniquely identifies each shipment.
For_Order [FK]	FK	Points to the Orders row this shipment fulfils.
Courier_Partner	-	The logistics company handling delivery.
Tracking_Number	-	The courier's reference number for tracking the shipment.
Dispatched_On	-	The date the shipment left the warehouse.
Expected_By	-	The estimated delivery date.
Delivery_Status	-	Preparing, In Transit, Delivered, or Not Applicable (e.g. for cancelled orders).

4.8 Returns & Reviews

Returns & Reviews is where a customer's post-sale feedback about a specific order and product is captured, whether that is a cancellation, a damage report, or a star review.

Attribute	Key	Usage / Purpose
Reference_Number [PK]	PK	Uniquely identifies each return or review entry.
Related_Order [FK]	FK	Points to the Orders row this entry concerns.
Related_Product [FK]	FK	Points to the Products row this entry concerns.
Raised_By [FK]	FK	Points to the Customers row that raised the entry.
Type	-	Cancellation, Damage Report, or Review.
Reason_Or_Feedback	-	Free-text explanation from the customer.
Rating	-	A 1-5 star score, populated only for reviews.
Status	-	Refund Issued, Under Review, or Published.

5. Entity-Relationship Diagram (Primary & Foreign Keys) DRAWIO

The diagram below shows all eight tables and how they connect, drawn in Chen notation with rectangles as entities, ovals as attributes (dashed ovals for derived attributes), and diamonds as relationships. Primary keys are underlined, and foreign keys are marked (FK), so that every relationship line can be traced directly to the key pair it connects — for example, Placed_By (FK) in Orders connects to Customer_Number, the primary key of Customers, through the PLACES relationship.

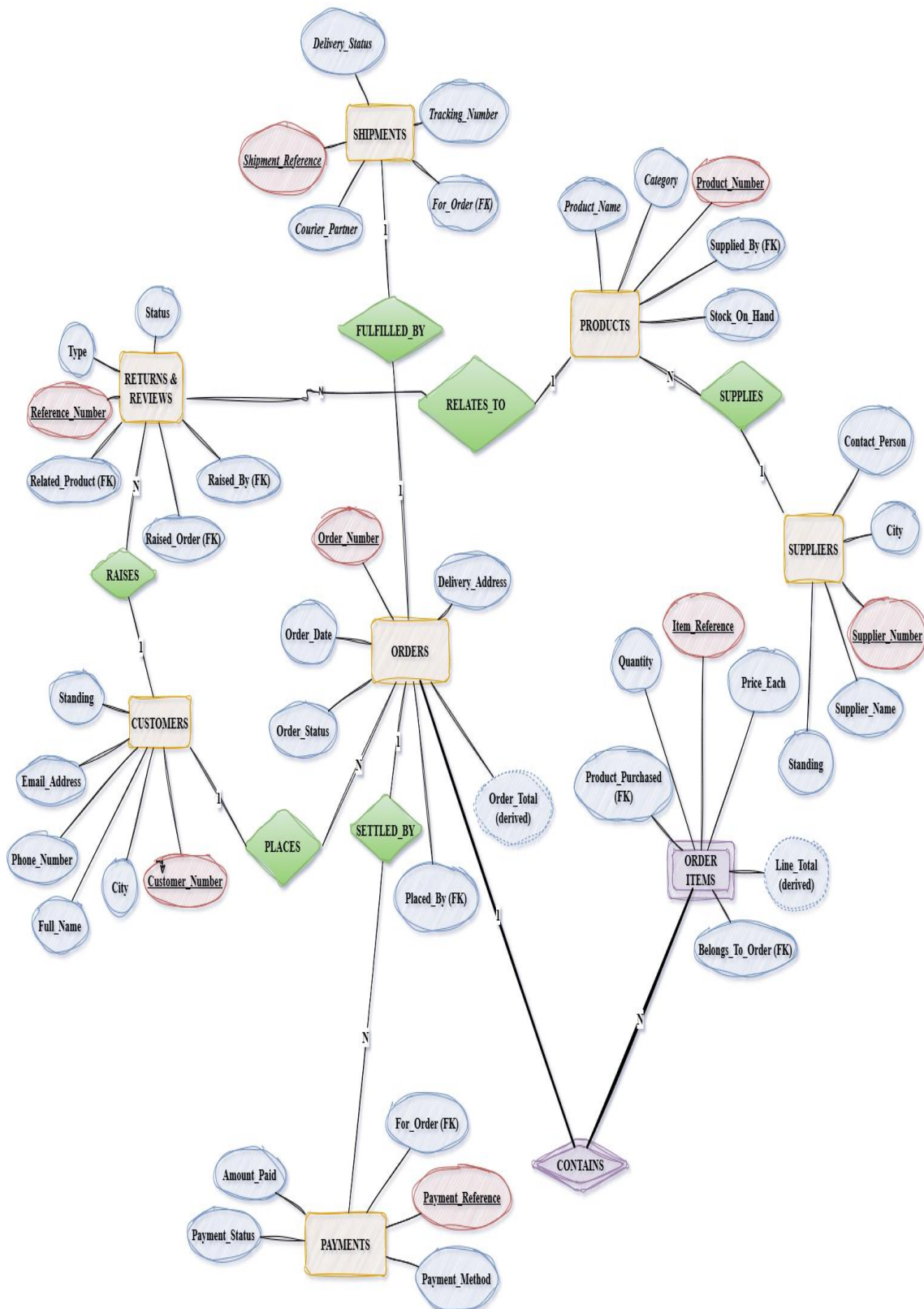
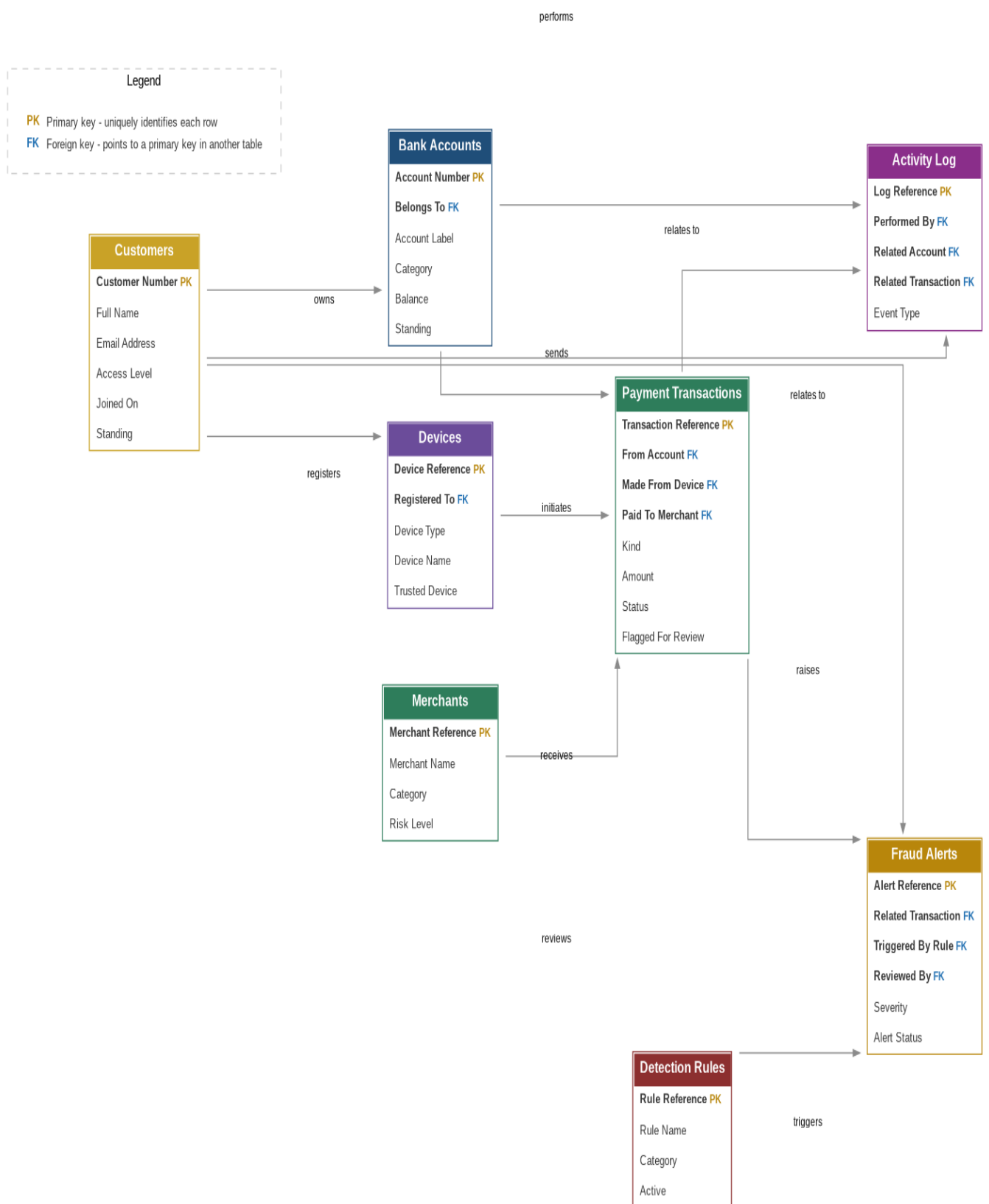


Figure 1: Product Order Management System — ER Diagram (Chen Notation)

Embedded Spreadsheet — Fraud Management Data Model



6. Spreadsheet Data Model — Sample Records

The full working spreadsheet behind this data model, Product_Order_Management_System.xlsx, is linked below so that every table, its sample records, and the cross-table lookup formulas can be viewed and edited directly. The draw.io source of the ER diagram is linked alongside it. For quick reference without opening either file, the same sample data is also reproduced as tables in the remainder of this section.

Spreadsheet (relational tables with PK/FK data):

https://docs.google.com/spreadsheets/d/196ZaJkvaapdjAw4KPJn4IBPh_XVduGe/edit?usp=drivesdk&ouid=117615247494676556422&rtpof=true&sd=true

Draw.io ER Diagram source (editable):

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6.1 Customers

Customer Number	Full Name	Email Address	Phone Number	City	Joined On	Standing
CUST-1001	Ananya Rao	ananya.rao@example.com	+91 98450 11223	Chennai	2023-02-11	Active
CUST-1002	Vikram Sethi	vikram.sethi@example.com	+91 98220 33445	Mumbai	2022-11-05	Active
CUST-1003	Meera Iyer	meera.iyer@example.com	+91 99010 55667	Bengaluru	2021-06-18	Active
CUST-1004	Rohan Das	rohan.das@example.com	+91 97170 77889	Delhi	2024-01-27	Inactive

6.2 Suppliers

Supplier Number	Supplier Name	Contact Person	Contact Email	City	Since	Standing
SUPP-9001	Nova Textiles	Rakesh Menon	rakesh@novatextiles.com	Coimbatore	2019-05-10	Active
SUPP-9002	BrightHome Goods	Priya Nambiar	priya@brighthouse.com	Ahmedabad	2020-08-22	Active
SUPP-9003	Everline Electronics	Suresh Kumar	suresh@everline.com	Pune	2018-03-14	Active
SUPP-9004	Urban Kitchenware	Divya Pillai	divya@urbankitchen.com	Hyderabad	2021-11-02	On Hold

6.3 Products

Product Number	Product Name	Supplied By	Category	Price (INR)	Stock	Listed On
PROD-3001	Cotton Bedsheet Set	SUPP-9001	Home & Living	1,499	120	2022-01-15
PROD-3002	Ceramic Dinner Set	SUPP-9002	Kitchen	2,299	65	2022-03-02
PROD-3003	Wireless Earbuds	SUPP-9003	Electronics	3,499	200	2023-06-19
PROD-3004	Non-Stick Pan	SUPP-9004	Kitchen	899	150	2023-09-05

6.4 Orders

Order Number	Placed By	Order Date	Delivery Address	Total (INR)	Status
ORD-5001	CUST-1001	2026-07-10	12 Lake View Road, Chennai	1,499	Delivered
ORD-5002	CUST-1002	2026-07-12	45 Marine Drive, Mumbai	5,798	Shipped
ORD-5003	CUST-1003	2026-07-15	78 MG Road, Bengaluru	3,499	Processing
ORD-5004	CUST-1004	2026-07-16	9 Connaught Place, Delhi	899	Cancelled

6.5 Order Items

Item Ref.	Belongs To Order	Product Purchased	Qty	Price Each	Line Total
ITEM-7001	ORD-5001	PROD-3001	1	1,499	1,499
ITEM-7002	ORD-5002	PROD-3002	1	2,299	2,299
ITEM-7003	ORD-5002	PROD-3004	1	3,499	3,499
ITEM-7004	ORD-5003	PROD-3003	1	3,499	3,499
ITEM-7005	ORD-5004	PROD-3004	1	899	899

6.6 Payments

Payment Ref.	For Order	Amount (INR)	Method	Paid On	Status
PAY-4001	ORD-5001	1,499	UPI	2026-07-10	Successful
PAY-4002	ORD-5002	5,798	Credit Card	2026-07-12	Successful
PAY-4003	ORD-5003	3,499	Net Banking	2026-07-15	Successful
PAY-4004	ORD-5004	899	UPI	2026-07-16	Refunded

6.7 Shipments

Shipment Ref.	For Order	Courier Partner	Tracking No.	Dispatched	Status
SHIP-6001	ORD-5001	Speedex Logistics	SPX88451209	2026-07-10	Delivered
SHIP-6002	ORD-5002	QuickRoute Couriers	QRC77213890	2026-07-13	In Transit
SHIP-6003	ORD-5003	Speedex Logistics	SPX88452317	2026-07-16	Preparing
SHIP-6004	ORD-5004	-	-	-	Not Applicable

6.8 Returns & Reviews

Reference	Related Order	Related Product	Type	Rating	Status
RTRN-8001	ORD-5004	PROD-3004	Cancellation	-	Refund Issued
RVW-8002	ORD-5001	PROD-3001	Review	5	Published
RVW-8003	ORD-5003	PROD-3003	Review	4	Published
RTRN-8004	ORD-5002	PROD-3002	Damage Report	-	Under Review

7. Additional Discussion: Applications, Advantages & Future Scope

7.1 Real-World Applications

- Retail and e-commerce platforms — tracking every order from cart to delivery across web, mobile, and marketplace channels.
- Warehouse and inventory management — using Stock_On_Hand and Supplied_By to trigger reordering from the correct supplier.
- Customer service and returns desks — using Returns & Reviews to resolve cancellations and damage claims against the exact order and product involved.

- Marketplace seller dashboards — adapting the same order/payment/shipment pattern to multi-vendor marketplaces.

7.2 Advantages of This Design

- Separating Order Items from Orders means an order can contain any number of products without repeating columns or altering the schema.
- Every payment and shipment carries a full audit trail back to the order, which supports customer support, refunds, and dispute resolution.
- Payments and Shipments progress through their own status values independently of Order_Status, so partial fulfilment or delayed payment can be represented without conflicting states.
- Returns & Reviews referencing both the order and the product keeps post-sale feedback traceable to the exact transaction, not just the product in general.

7.3 Future Enhancements

- Introducing a Discounts/Coupons table linked to Orders to support promotional pricing.
- Adding a Warehouses table and linking it to Products for multi-location stock tracking.
- Extending Shipments to support multiple partial shipments per order.
- Extending Payments to support multiple part-payments per order.
- Adding a Notifications table to track how and when a customer was informed about their order or return status.

8. Conclusion

This assessment has presented a complete, eight-table data model for a Product Order Management System, along with an entity-relationship diagram marking every primary and foreign key, and the underlying spreadsheet made directly accessible within this document. The design demonstrates how a real retail order management workflow — from a customer placing an order, through payment and shipment, to post-sale returns and reviews — can be represented as a normalised relational schema. The model is extensible, and Section 7 outlines several directions in which it could be developed further for production use.

9. References

- VIT Vellore, School of Computer Science and Engineering — Database Management Systems course material.
- Chen, P. P. (1976). The Entity-Relationship Model — Toward a Unified View of Data. ACM Transactions on Database Systems.
- Product_Order_Management_System.xlsx — project spreadsheet, linked in Section 6 of this report.

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